MENT OF ELECTRONICS &COMMUNICATION ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

SUBJECT :SIGNAL AND SYSTEMS

Class: BE (CSE)
Semester: IV
Academic Year: 2025-26

Max Marks: 30
Duration: 90 Min
Date : 27-06-2025

	PART A Answer All Questions (10 M)	Marks	BT	CO
1. a)	State and prove the integration property of laplace transforms	2	1	4
b)	What is meant by the ROC of the z-transform of a sequence x(n)	2	2	5
c)	Derive the relationship between laplace transform and z transform	2	3	5
d	What are Dirichlet conditions of Fourier transforms	2	4	4
e	Findout DTFT u(n-m)	2	1	5
	PART B Answer Any Two Questions (20 M)			
2. a)	Findout Inverse Laplace transform of a sequence x(s) i) $X(s)=\frac{2s+1}{(s+1)(s^2+2s+2)}$ ii) $X(s)=\frac{4s^2+8s+10}{(s+2)(s^2+2s+5)}$.	8	2	4
b	Find Fourier tranform of Rectangular pulse	2	5	4
3. a)	Find the Inverse z transform of $X(z)=\frac{(3+\frac{11}{2}z^{-1}+7z^{-2})}{(1-\frac{1}{2}z^{-1})(1+2z^{-1}+4z^{-2})}$ using Partial fraction expansion method.	6	3	5
b	State and prove Time shifting ~~and~~ Time convolution property the DTFT	4		5
4. a)	State and prove Time scalling and Parseval's theorem of the Fourier transform.	4	4	4
b)	Realize the system with following Transfer-function $X(s)=\frac{s+2}{s^2+8s+25}$	6	5	5

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$\int_{-\infty}^{\infty} |x(t)|\,dt < \infty$

finite

$\sum x(n) z^{-n}$

$s = \sigma + j\omega$

$\text{sinc}(x) = \frac{\sin x}{x}$

$s+2+2$

$s^2+4s+4+21 = (s+2)^2+5^2$

$\frac{Ax+B}{(x+a)(x^2+bx+c)} = \frac{A}{x+a} + \frac{Bx+C}{x^2+bx+c}$

$\frac{A}{s+1} + \frac{Bs+C}{s^2+2s+2}$

$As^2+2As+2A+Bs^2+Cs+$

$s^2(A+B)+s(2$

$x(t)e^{-st}$

$\frac{1}{j\omega}$

$A\cos(\omega_0 t+\phi)$

$|\cos(\omega_0 t+\phi+c$